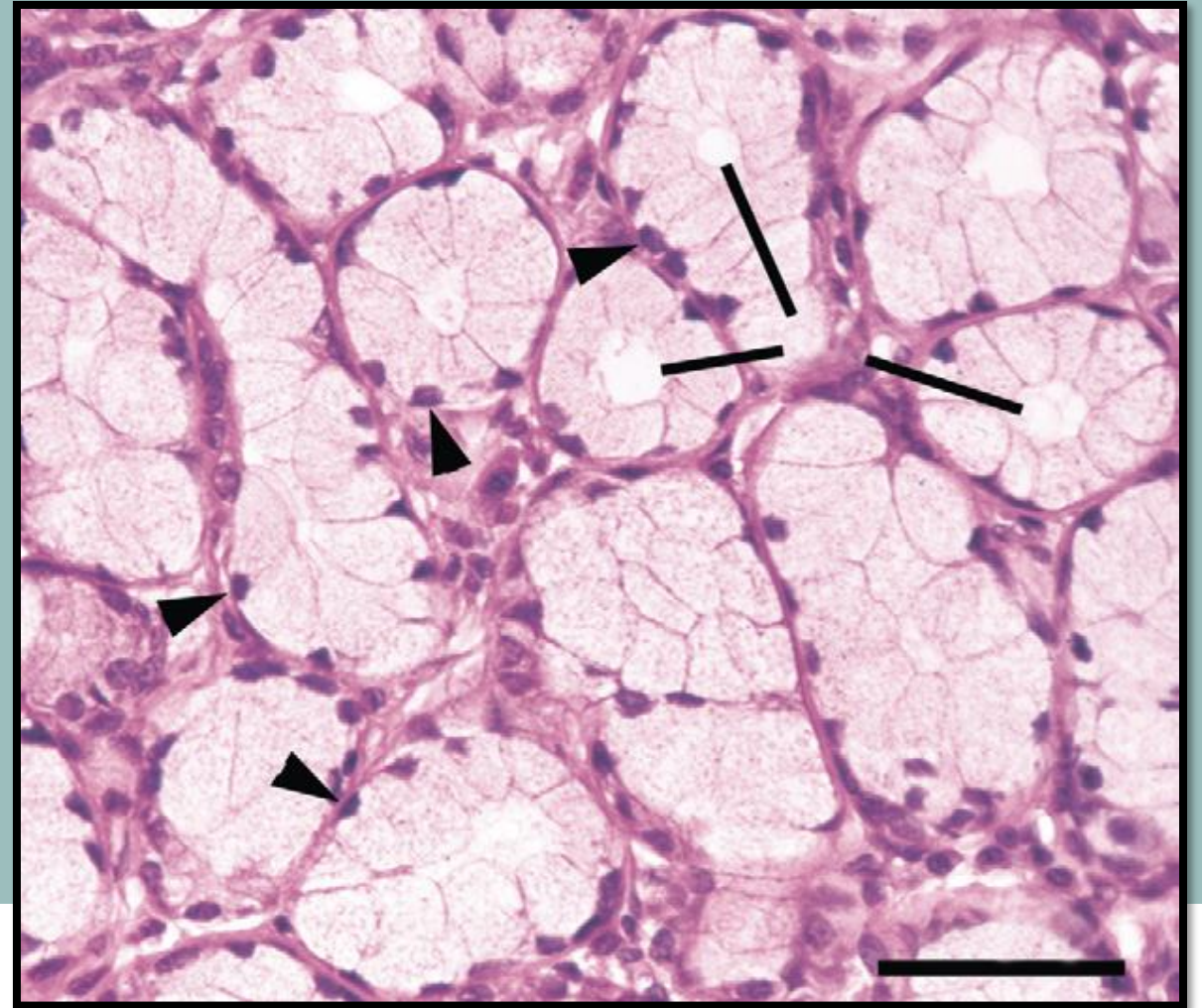




SALIVARY GLANDS TUTORIAL

Dr. Misbah ashraf moten

**TASK 1: IDENTIFY, PASTE, STUDY AND
LABEL THE FOLLOWING FIGURES**

FIG. 16.2 THE SECRETORY AND DUCTAL ELEMENTS IN A MIXED SALIVARY GLAND.

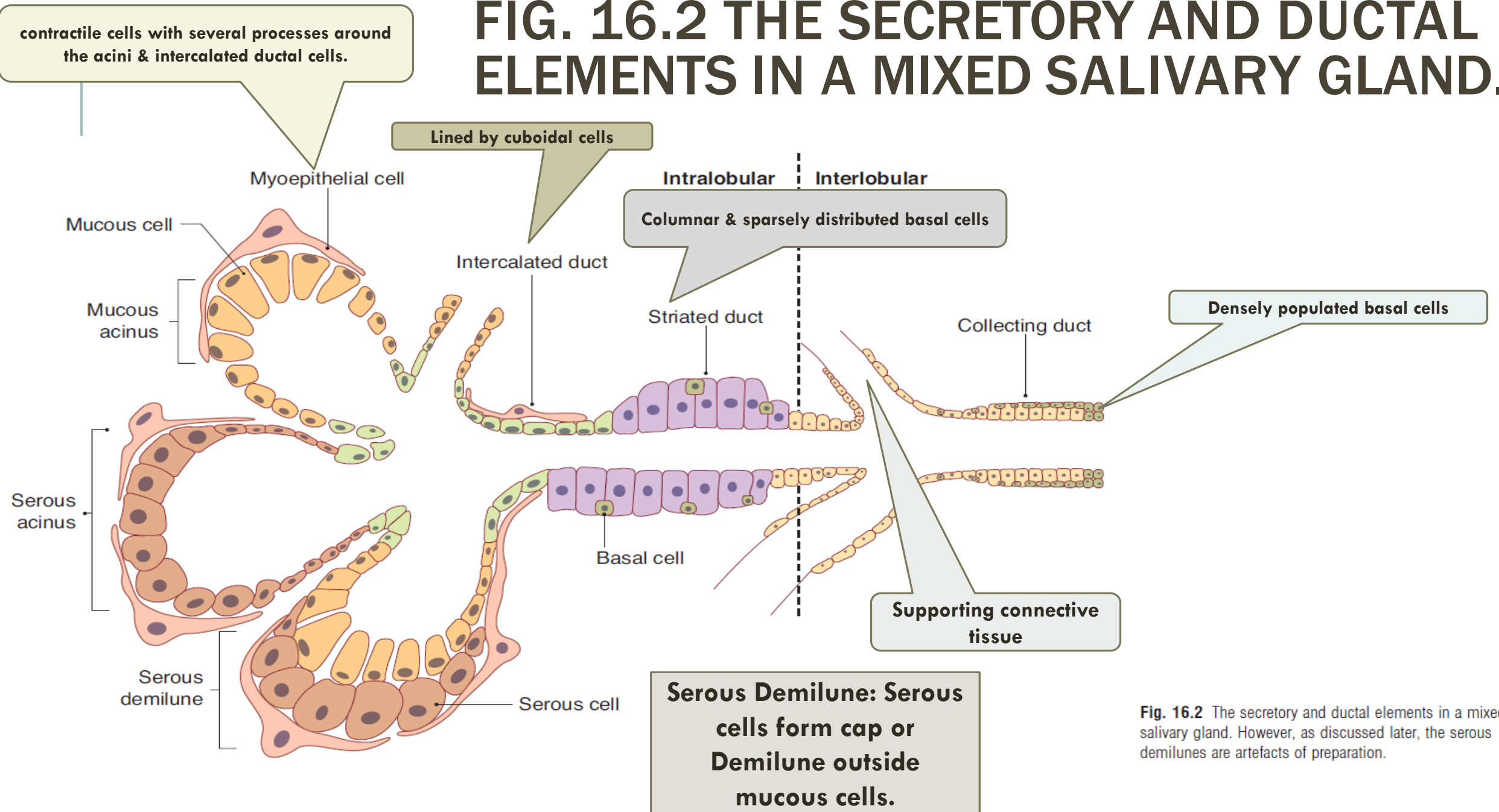


Fig. 16.2 The secretory and ductal elements in a mixed salivary gland. However, as discussed later, the serous demilunes are artefacts of preparation.

FIG. 16.1. GENERAL REPRESENTATION OF A SALIVARY GLAND

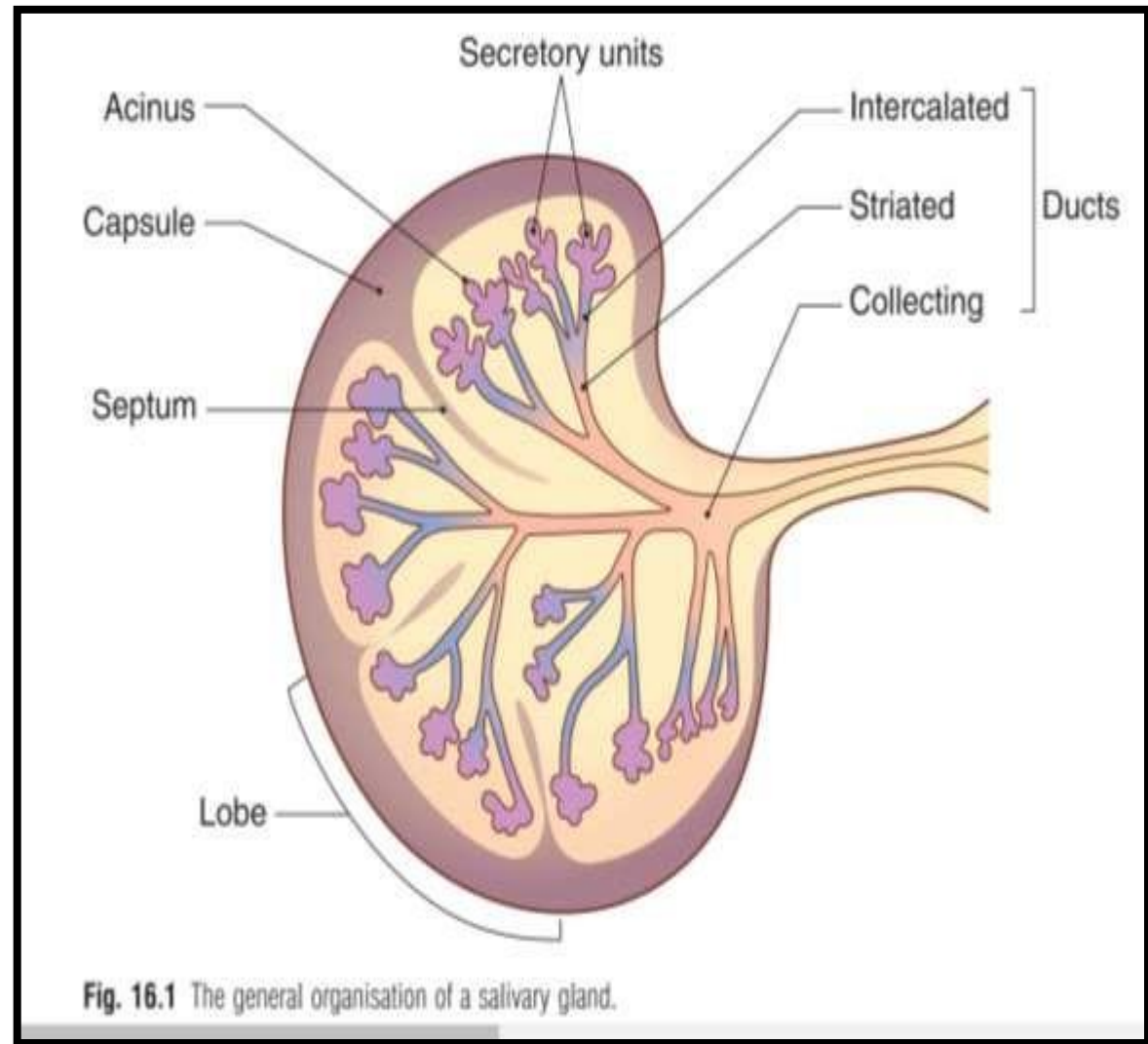


FIG. 16.6 DARK, GRANULAR APPEARANCE OF CYTOPLASM OF SEROUS ACINI OF PAROTID GLAND:

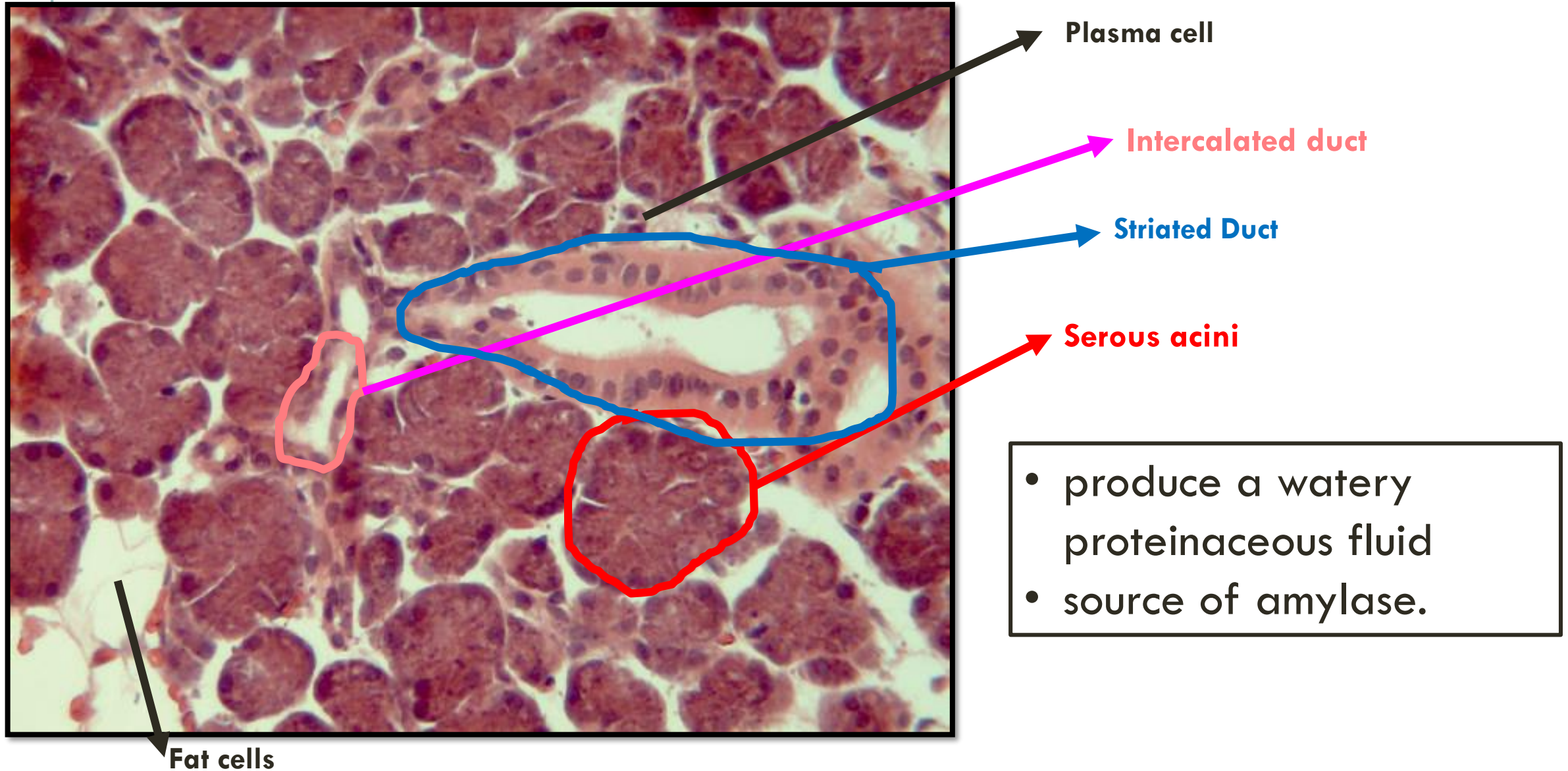
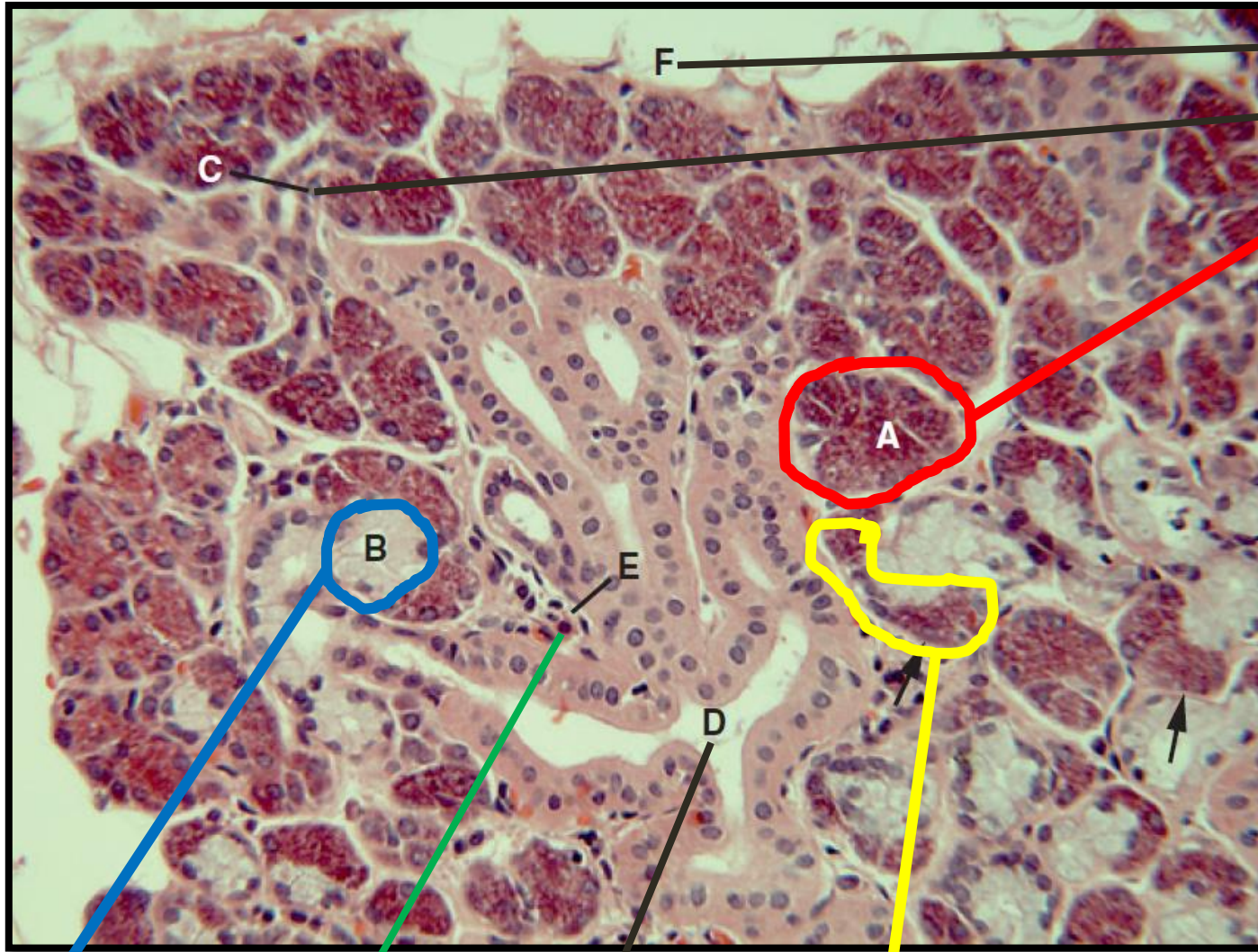


FIG. 16.26: SUBMANDIBULAR GLAND SHOWING DARKER SEROUS (A) AND LIGHTER MUCOUS (B) ACINI



Fat cells

Intercalated
duct

Serous acini

Crescent-shaped serous cells form
a cap or demilune, outside the
mucous cells

Plasma Cell

Serous Demilune

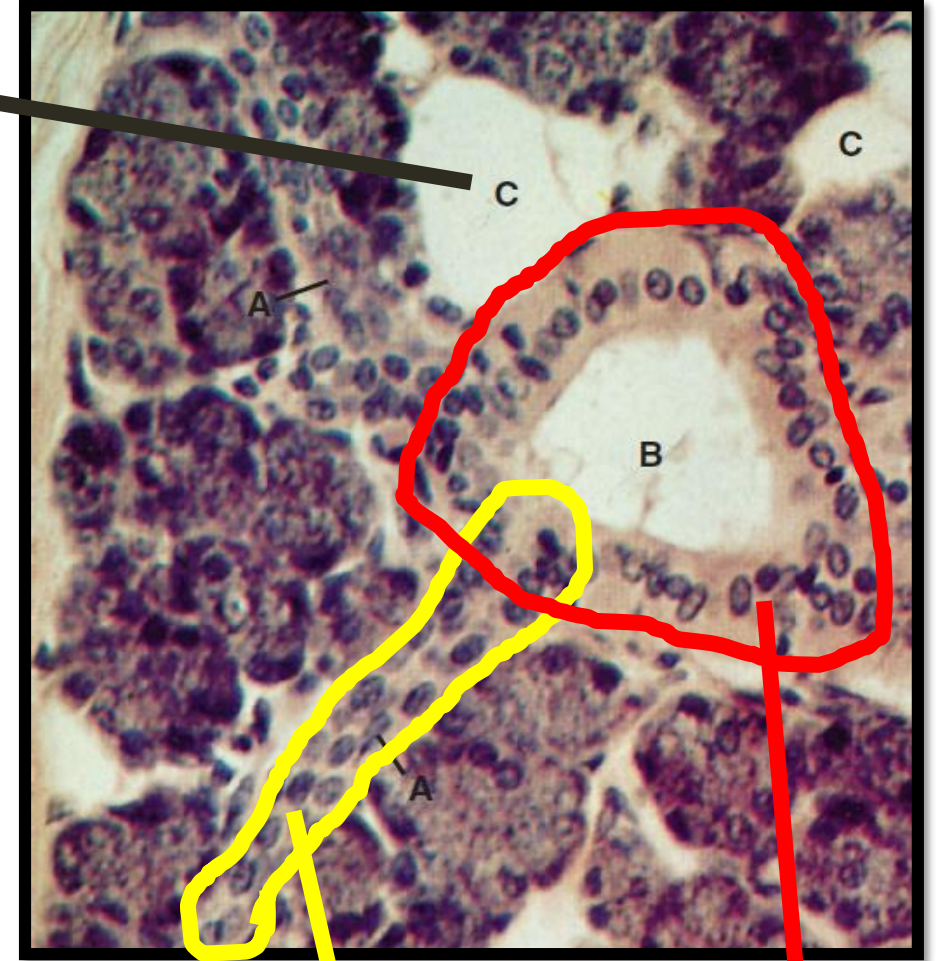
Striated duct

Mucous acini

FIG. 16.17 PAROTID GLAND SHOWING INTERCALATED DUCTS (A) LEADING FROM SEROUS ACINI AND INTO THE STRIATED DUCT (B).

- The cuboidal epithelial cells lining the intercalated ducts have prominent nuclei due to their scanty cytoplasm.
- The nuclei in the striated ducts are more centrally positioned within the cells

Fat cell



Intercalated ducts

Striated ducts

MODIFICATION OF SALIVA

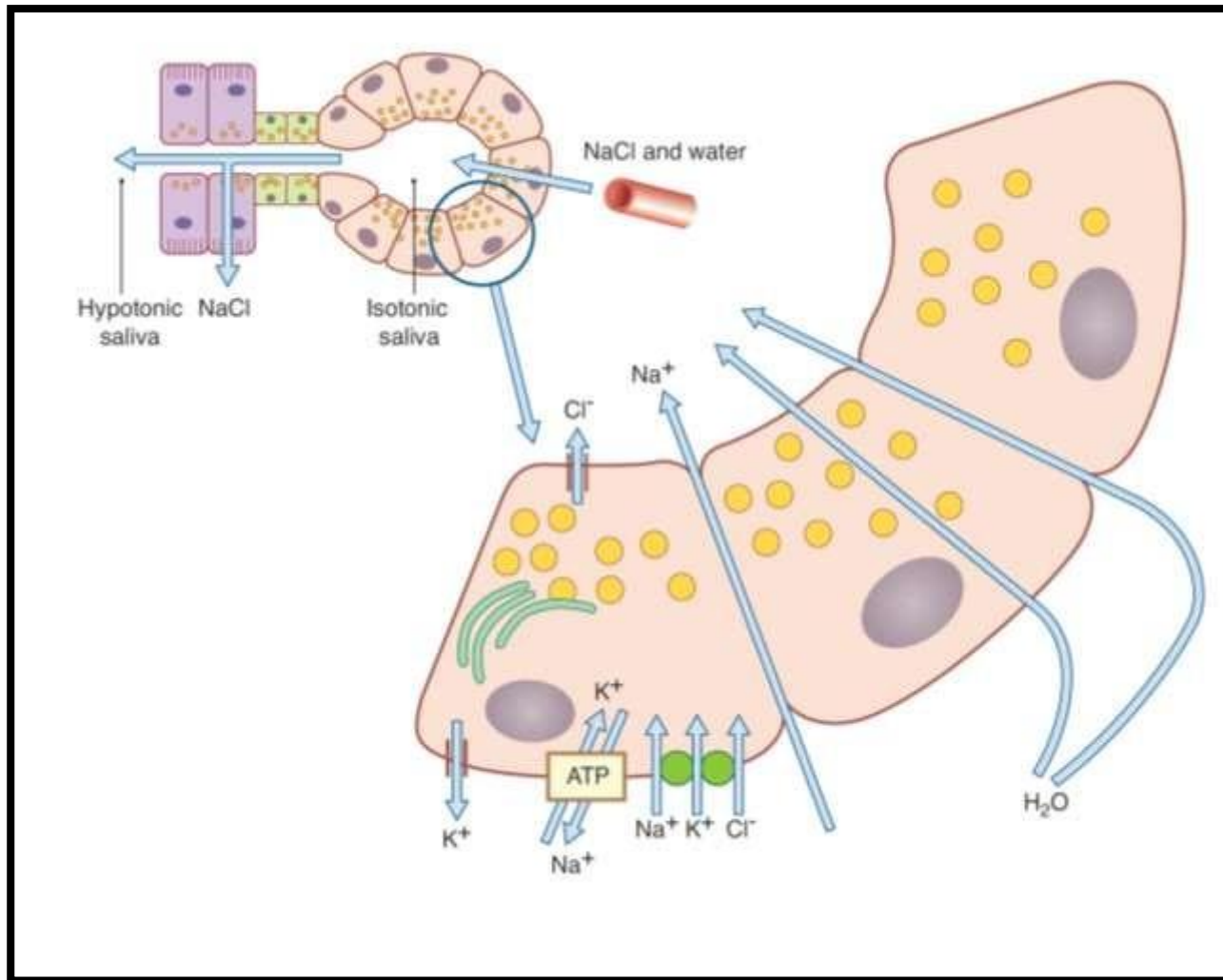


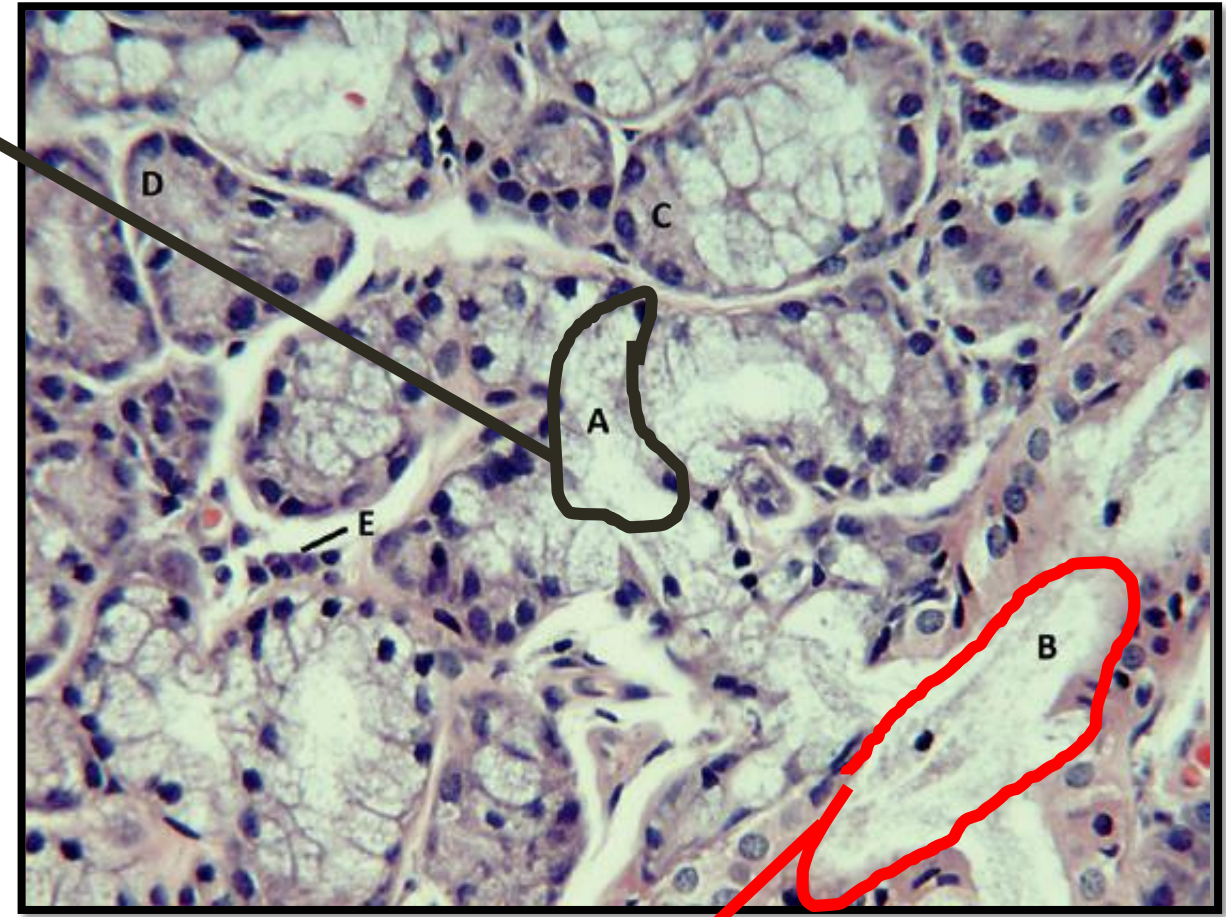
Fig. 16.16: Salivary fluid secretion by acinar cells and the resorption of salt in striated duct cells.

FIG. 16.37 PALE STAINING MUCOUS ACINI (A) OF SUBLINGUAL GLAND DRAIN INTO A COLLECTING DUCT (B).

Mucous acini

The appearance of serous demilunes (C) & acini (D) is caused by the scarcity of secretory granules in mucous cells at the beginning of the secretory cycle, although parts of these cells show pale areas as secretory granules accumulate.

Plasma cells are seen interstitially (E)



Collecting Ducts

TASK 2: ANSWER THE FOLLOWING QUESTIONS

Q1. RELATE THE COMPOSITION OF SALIVA TO IMPORTANT FUNCTIONS:

Component	Major Constituents	Important Functions
Water	H ₂ O	Moistening, taste, swallowing
Mucins	Mucin glycoproteins	Lubrication, mucosal protection, pellicle formation
Minerals / Ions	Calcium, phosphate, bicarbonate	Buffering, remineralization, enamel protection
Digestive Enzymes	Amylase	Carbohydrate digestion
Antibacterial Proteins	Lysozyme, histatin, lactoferrin, defensins, cystatin	Antibacterial defense
Immunoglobulins	IgA	Immune protection
Growth Factors	EGF, NGF	Wound healing, epithelial maintenance
Taste-related Proteins	Gustin	Taste perception
Mineralization Regulators	Statherin	Prevents unwanted mineralization
Other Components	Kallikrein, blood group substances	Regulatory/protective roles

Q2. WHAT ARE THE FUNCTIONS OF MYOEPIHELAL CELLS FUNCTIONS?

Box 16.2 Myoepithelial activity within salivary glands

It can:

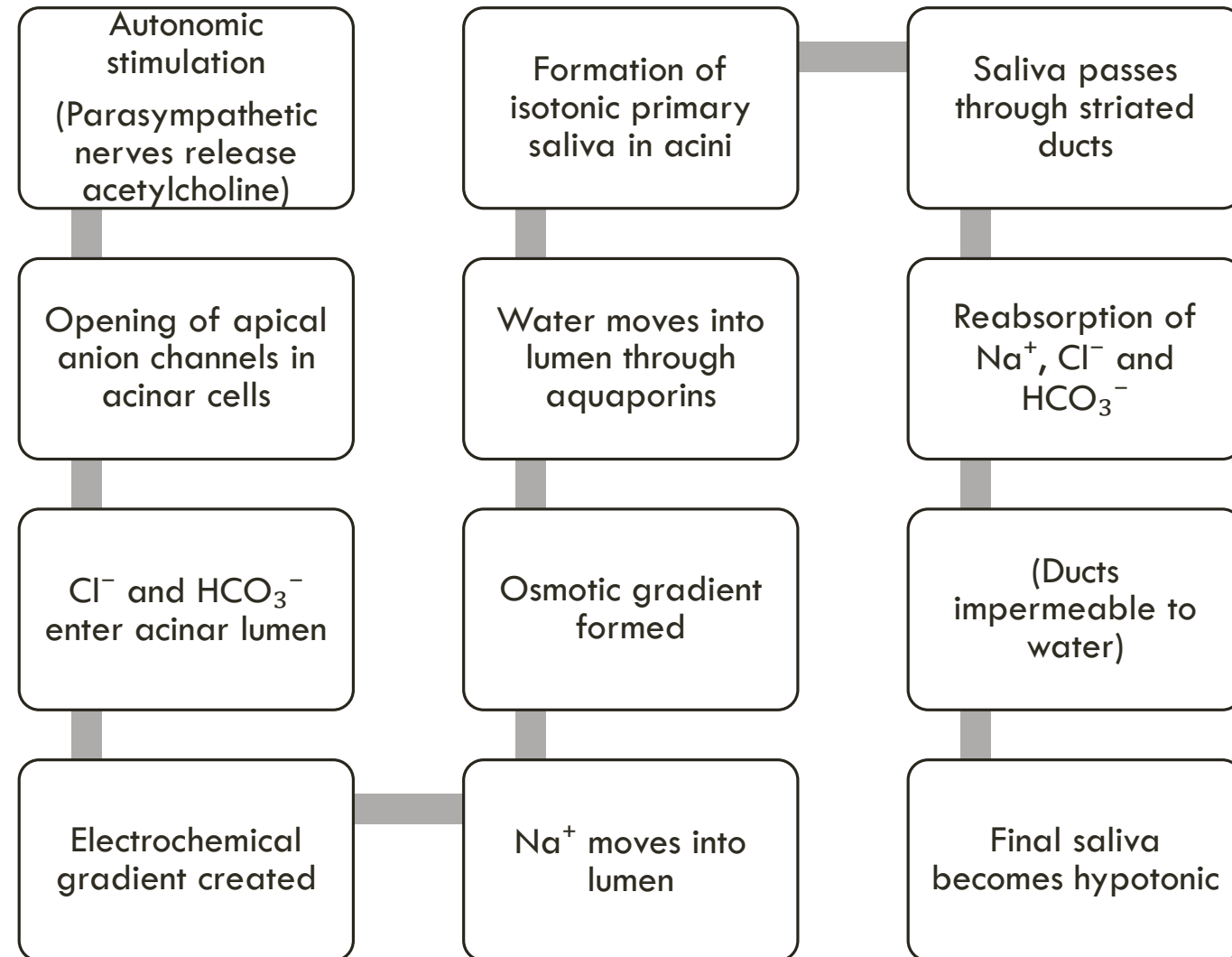
- support the underlying parenchyma and reduce back-permeation of fluid
- accelerate the initial outflow of saliva
- reduce luminal volume
- contribute to the secretory pressure
- help salivary flow to overcome increases in peripheral resistance – but if this is excessive it may lead to sialectatic damage of striated ducts, thereby increasing overall permeability.

Q3. DIFFERENTIATE BETWEEN THE MAJOR SALIVARY GLANDS

Parotid gland	Submandibular gland	Sublingual gland
Largest	Intermediate	Smallest
Serous	Mixed, mainly serous	Mucous
Intercalated ducts	Intercalated ducts	Few intercalated ducts
Striated ducts present	Striated ducts present	No striated ducts
Collecting ducts end in single main duct	Collecting ducts end in single main duct	Collecting ducts end in many main ducts
Nonspontaneous secretor	Nonspontaneous secretor	Spontaneous secretor
Saliva hypotonic	Saliva hypotonic	Saliva isotonic

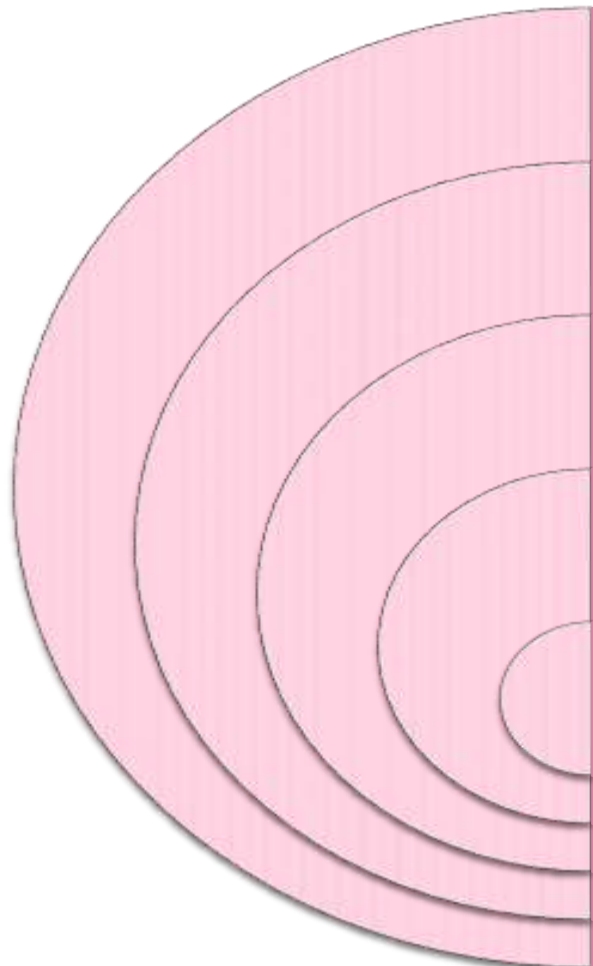
Q4. EXPLAIN THE PROCESS OF SALIVARY FORMATION & MODIFICATION

Creation of a **salt-secreting epithelium** by salivary acinar cells as Na^+ , Cl^- , HCO_3^- , & other ions a transportation into acinar lumen & creation of hypertonic environment compared with surrounding blood capillaries & interstitial fluid.



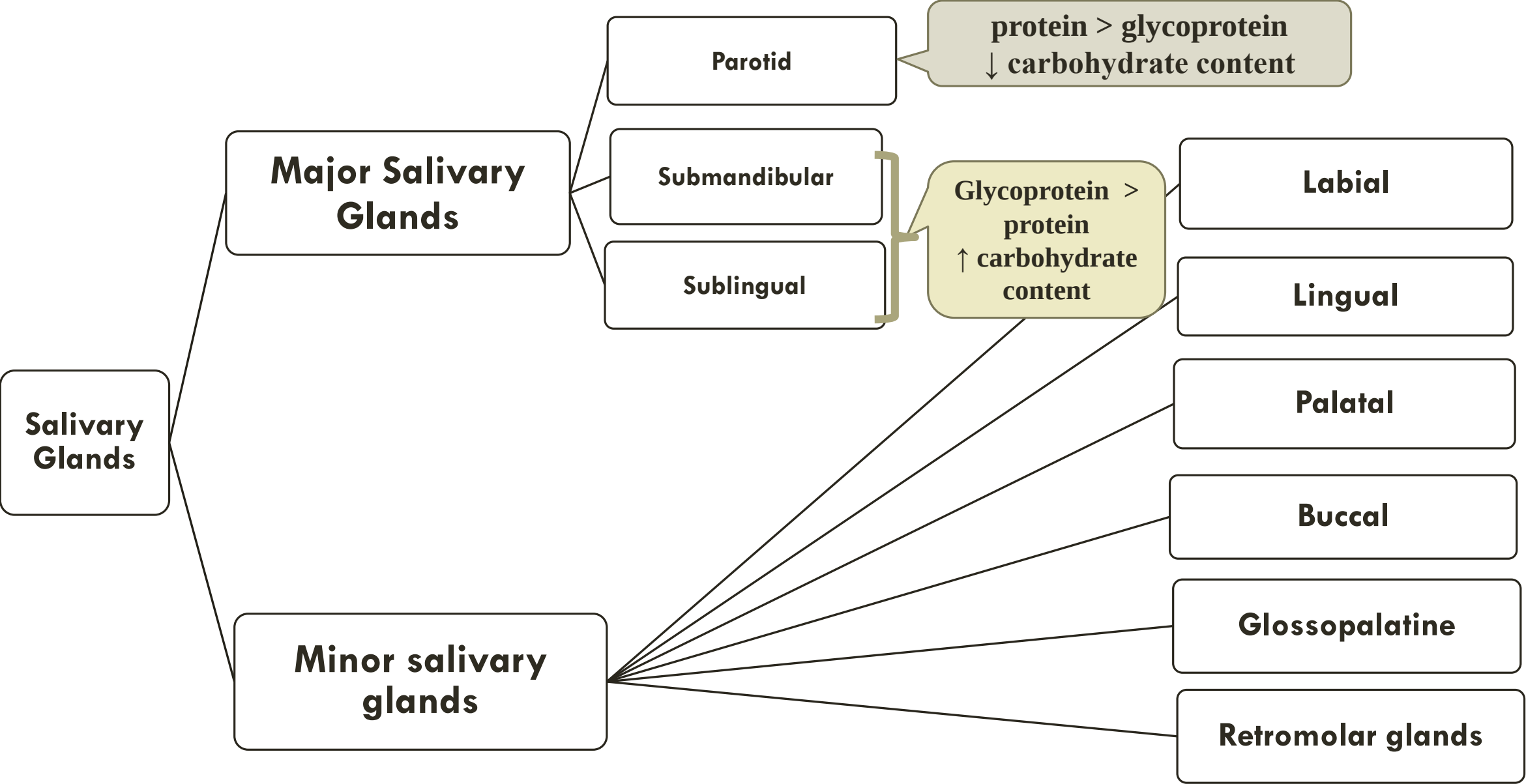
**IMPORTANT STUDY MATERIAL FOR
REFERENCE**

PRIMARY SALIVA

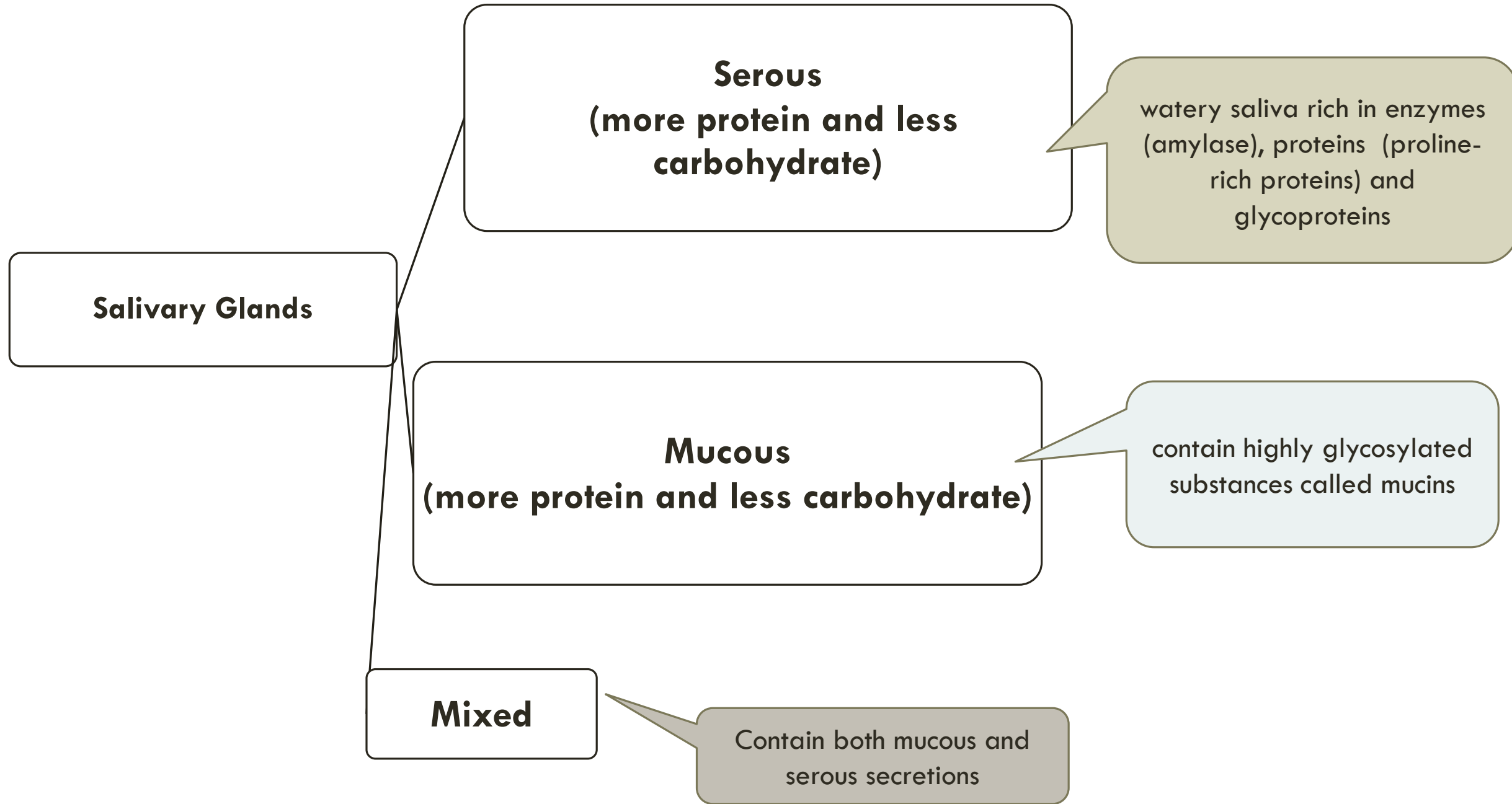


Salivary secretions controlled by ANS.
Nerve mediated secretion carried by parotid and submandibular gland
Spontaneous secretion carried by sublingual and minor salivary glands
0.3ml/min flowrate
500-750ml secretion per day (90% from major Salivary Glands)

CLASSIFICATION OF SALIVARY GLANDS ACCORDING TO SIZE



CLASSIFICATION OF SALIVARY GLANDS ACCORDING TO SECRETION



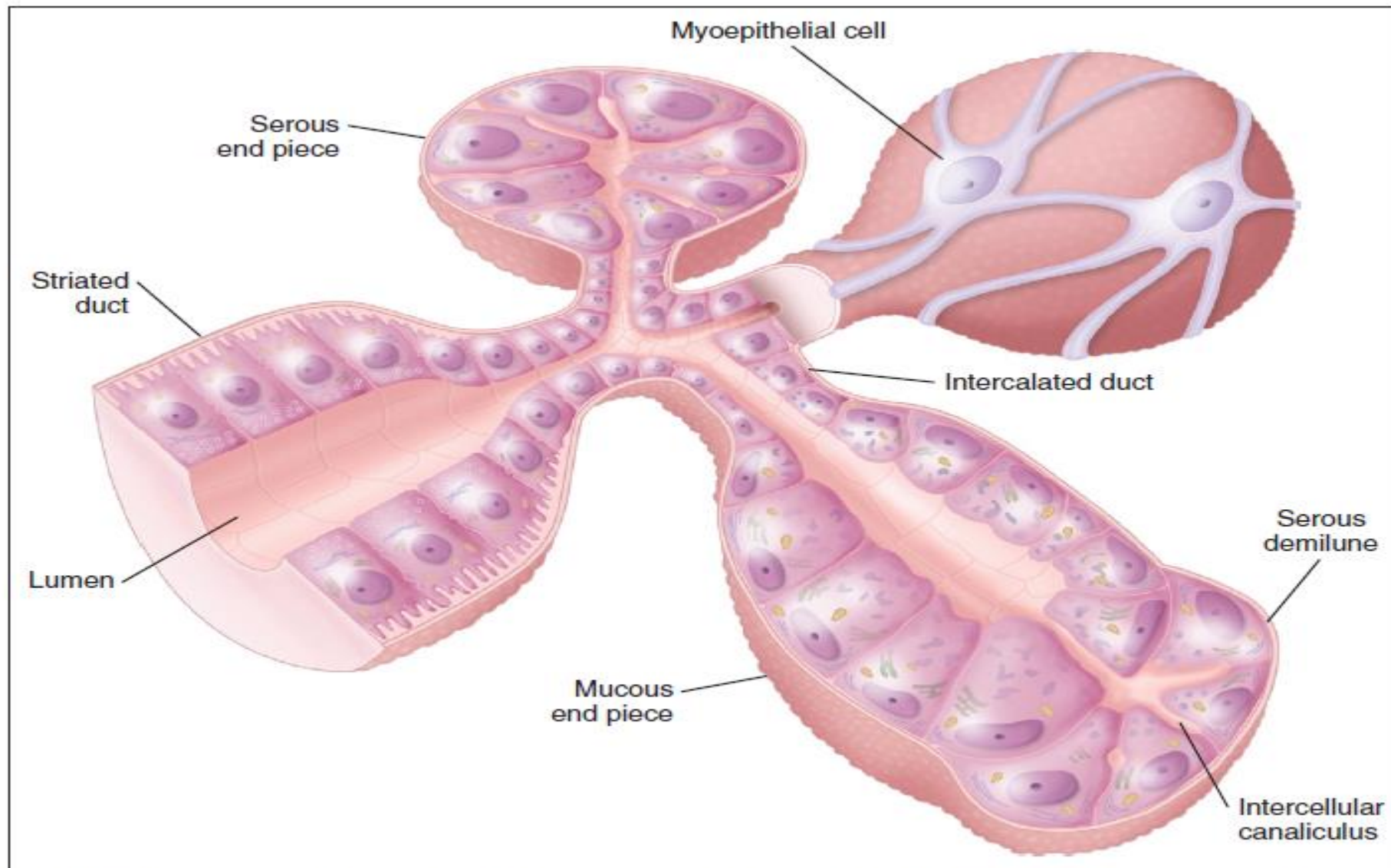
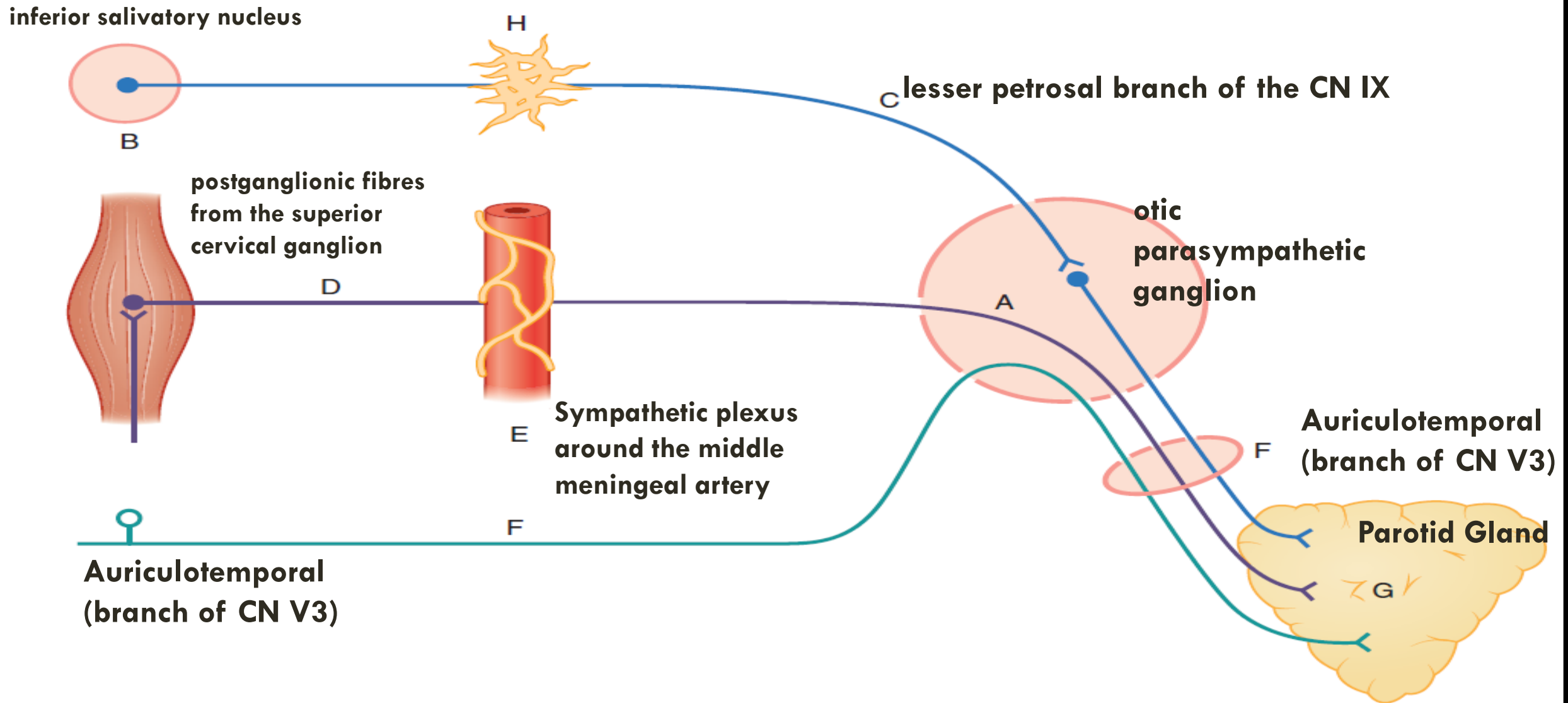


FIGURE 11-33 Architecture of salivary gland ducts and secretory end pieces and the main features of the parenchymal cells. (From Hand AR. In Bhaskar SN, editor: *Orban's oral histology and embryology*, ed 11, St Louis, 1991, Mosby.)

Fig. 4.13 THE SECRETOMOTOR SUPPLY TO THE PAROTID SALIVARY GLAND.



SECRETOMOTOR SUPPLY OF PAROTID GLAND

Origin: Parasympathetic fibers start in the **salivatory nucleus** (medulla oblongata).

Nerve pathway: Travel via the **glossopharyngeal nerve (CN IX)**.

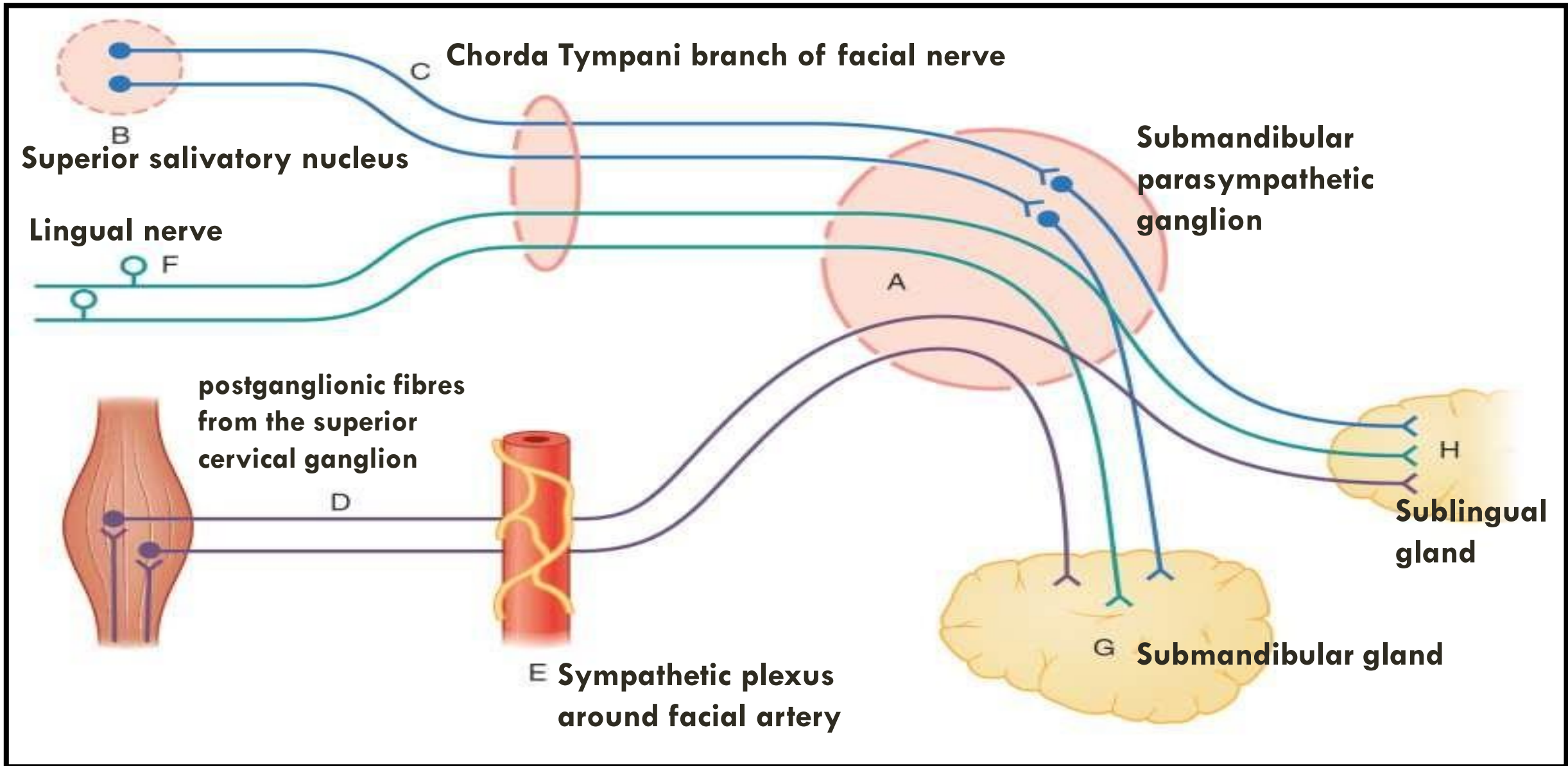
Synapse: Fibers synapse at the **otic ganglion** (near the ear).

Postganglionic route: Fibers go directly to the **parotid gland**.

Action: Release **acetylcholine** onto acinar cells → stimulates saliva secretion into the oral cavity.

Function: Responsible for **reflexive/anticipatory** salivation (triggered by sight, smell, thought, or taste of food).

FIG. 4.14 THE SECRETOMOTOR SUPPLY TO THE SUBMANDIBULAR AND SUBLINGUAL SALIVARY GLANDS.



SECRETOMOTOR SUPPLY OF SUBMANDIBULAR GLAND

Origin: Preganglionic fibers start in the **superior salivatory nucleus** (pons).

Nerve pathway: Fibers travel via the **facial nerve (CN VII)**.

Synapse: Fibers leave CN VII and synapse at the **submandibular ganglion** (near the gland).

Postganglionic route: Fibers from the ganglion travel along **lingual nerve** branches (from CN V3).

Neurotransmitter: Postganglionic fibers release **acetylcholine** onto acinar cells.

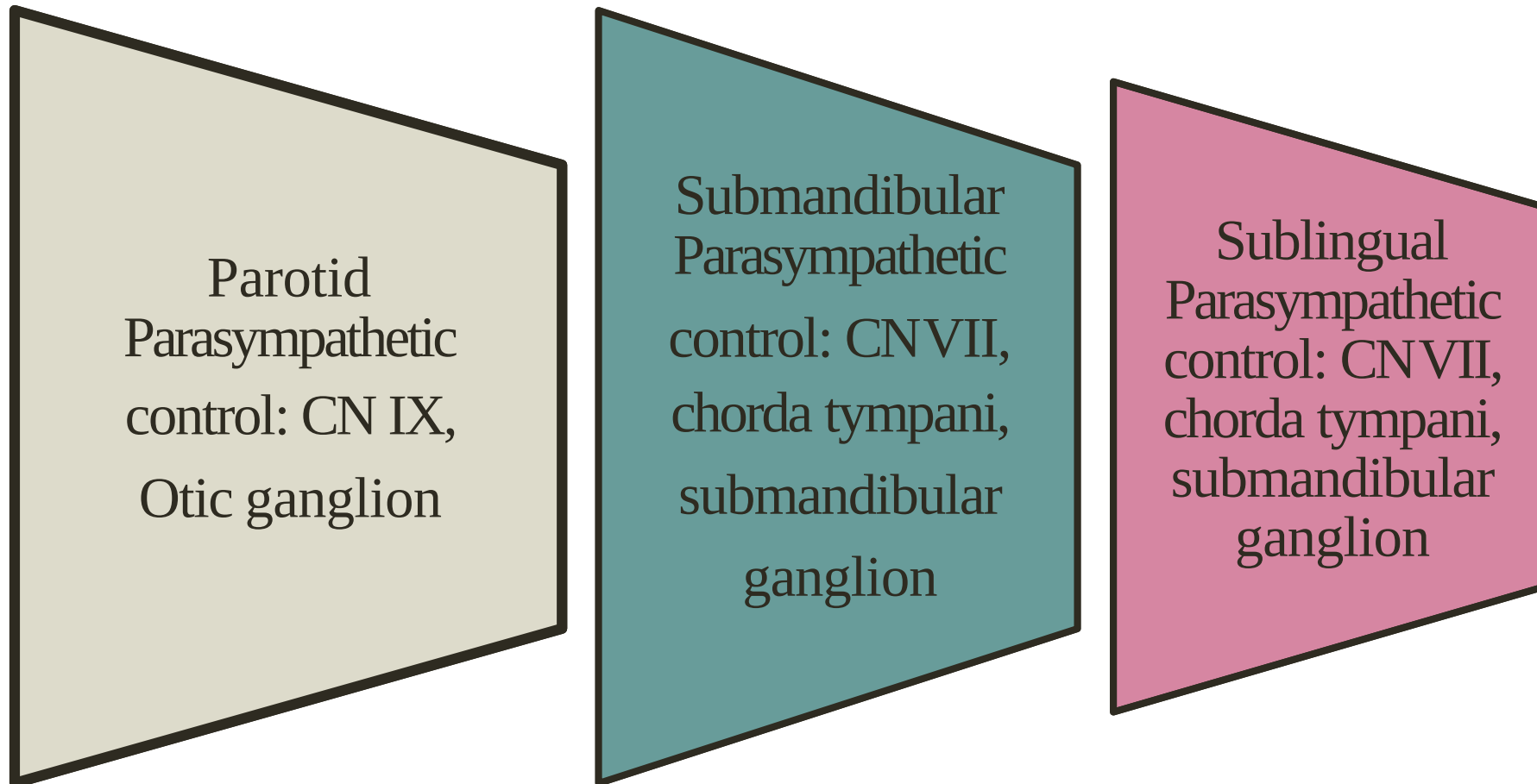
Effect: Stimulates saliva secretion into gland ducts.

Exit: Saliva drains via **Wharton's duct** into the floor of the mouth.

Note: The **sublingual gland** follows a similar parasympathetic pathway but with different ganglia/nerves.

Nerve supply of Major Salivary gland:

CN V, Sympathetic control: superior cervical ganglion



MYOEPIITHELIAL (BASKET) CELLS:

Shape around acini

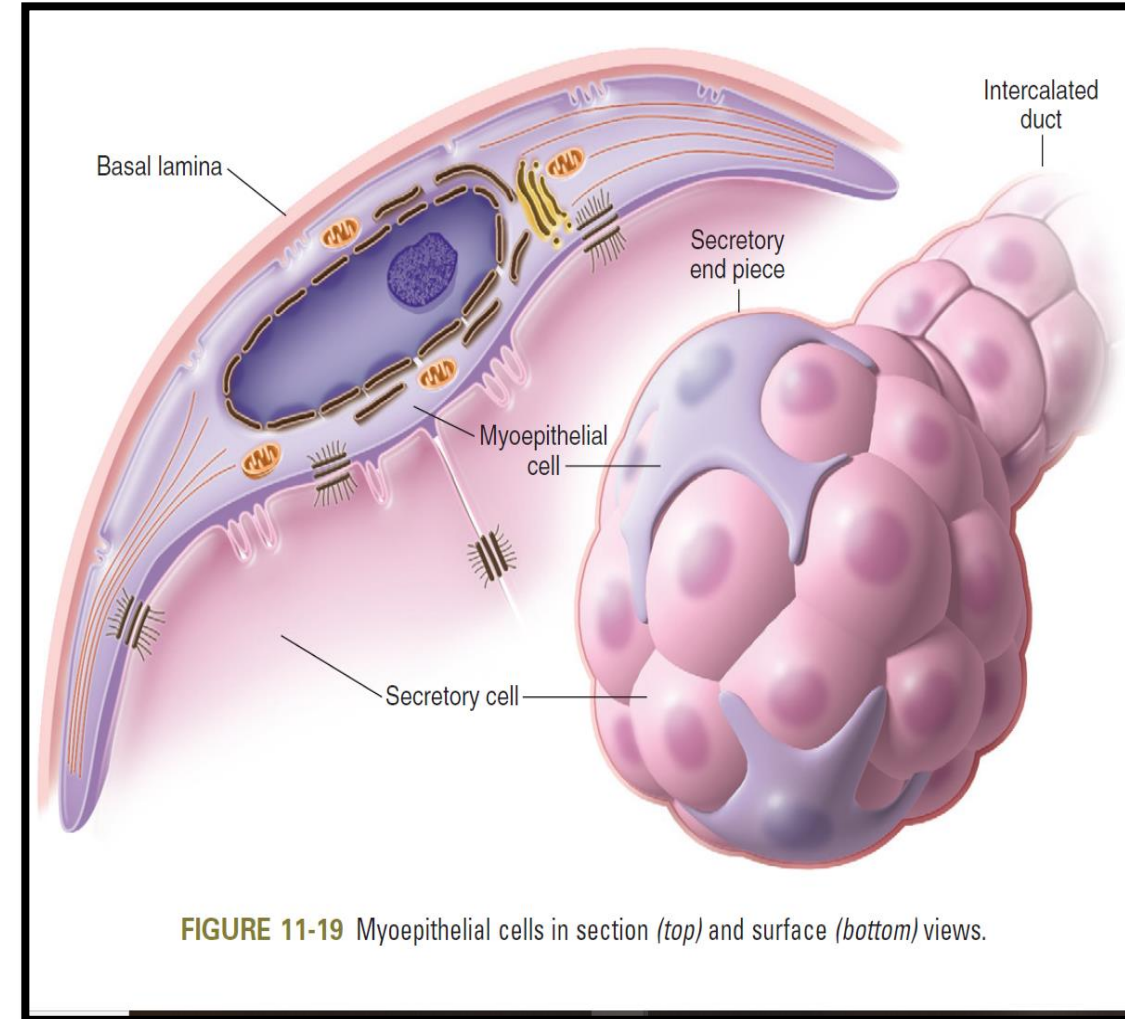
Dendritic cells, stellate shaped body

Some tapering processes lying in gutters on secretory cell surface, so acinar outline → smooth

Shape around intercalated ducts

Elongated cells, lying more superficially & running longitudinally along the duct

Produce a bulge in the outline of the duct & have few short processes



NEUROEFFECTOR SITES:

Neuro effectors are the target cells that respond to neurotransmitter signals, resulting in a physiological change.

Hypolemmal: Some neuroeffector sites occur beneath the basal lamina in direct contact with the plasma membrane

Epilemmal: Neuroeffector sites that remain outside the basal lamina – epilemmal

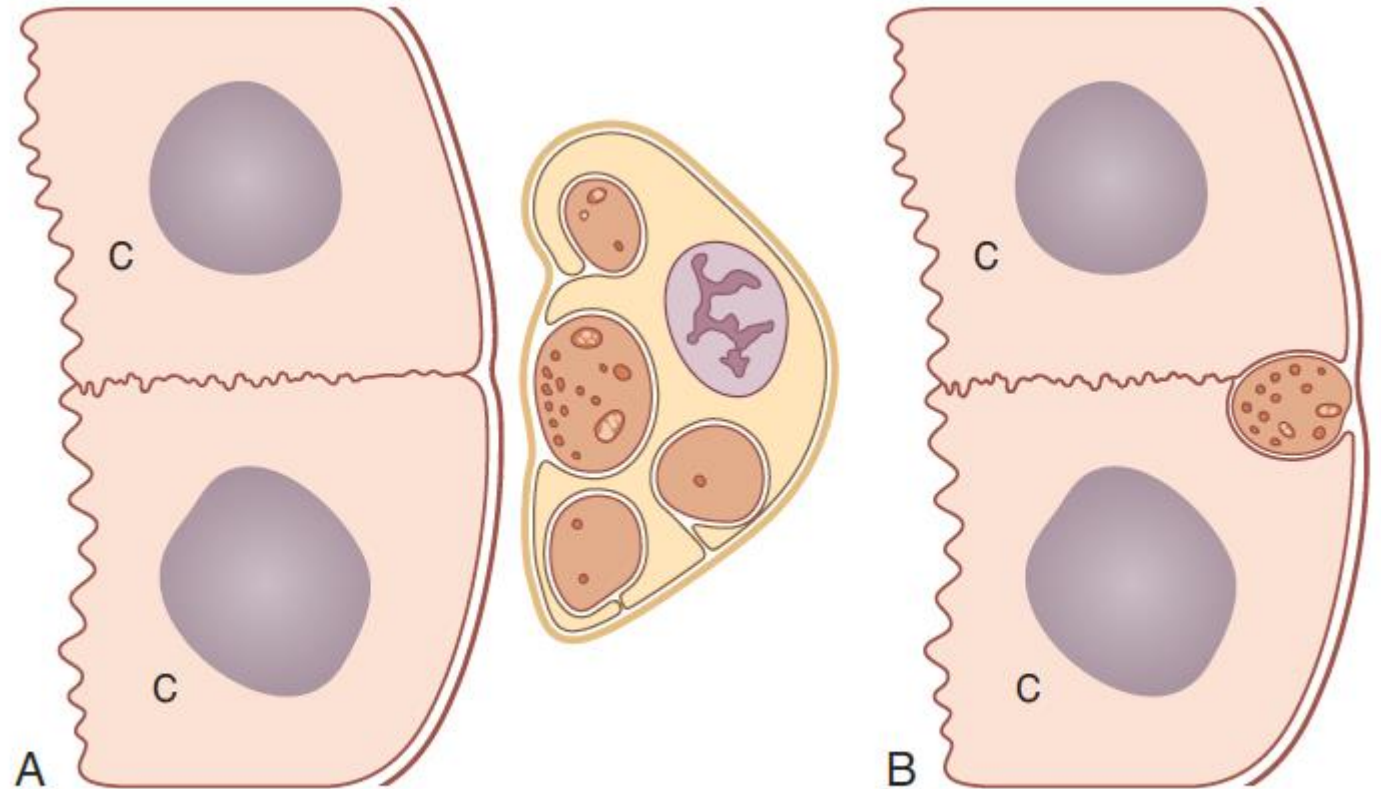


Fig. 16.12 Neuroeffector relationships in salivary glands. A = epilemmal; B = hypolemmal; C = parenchymal cells. Courtesy Prof. J R Garrett and Karger Press.

METHODS OF SALIVARY SECRETION

Salivary acinar cells secrete proteins by three mechanisms:

1. Regulated pathway (stored granule exocytosis)

- Main method of secretion.
- Proteins are stored in secretory granules and released after neuronal stimulation.
- Time from synthesis to secretion is about 3.5 hours.

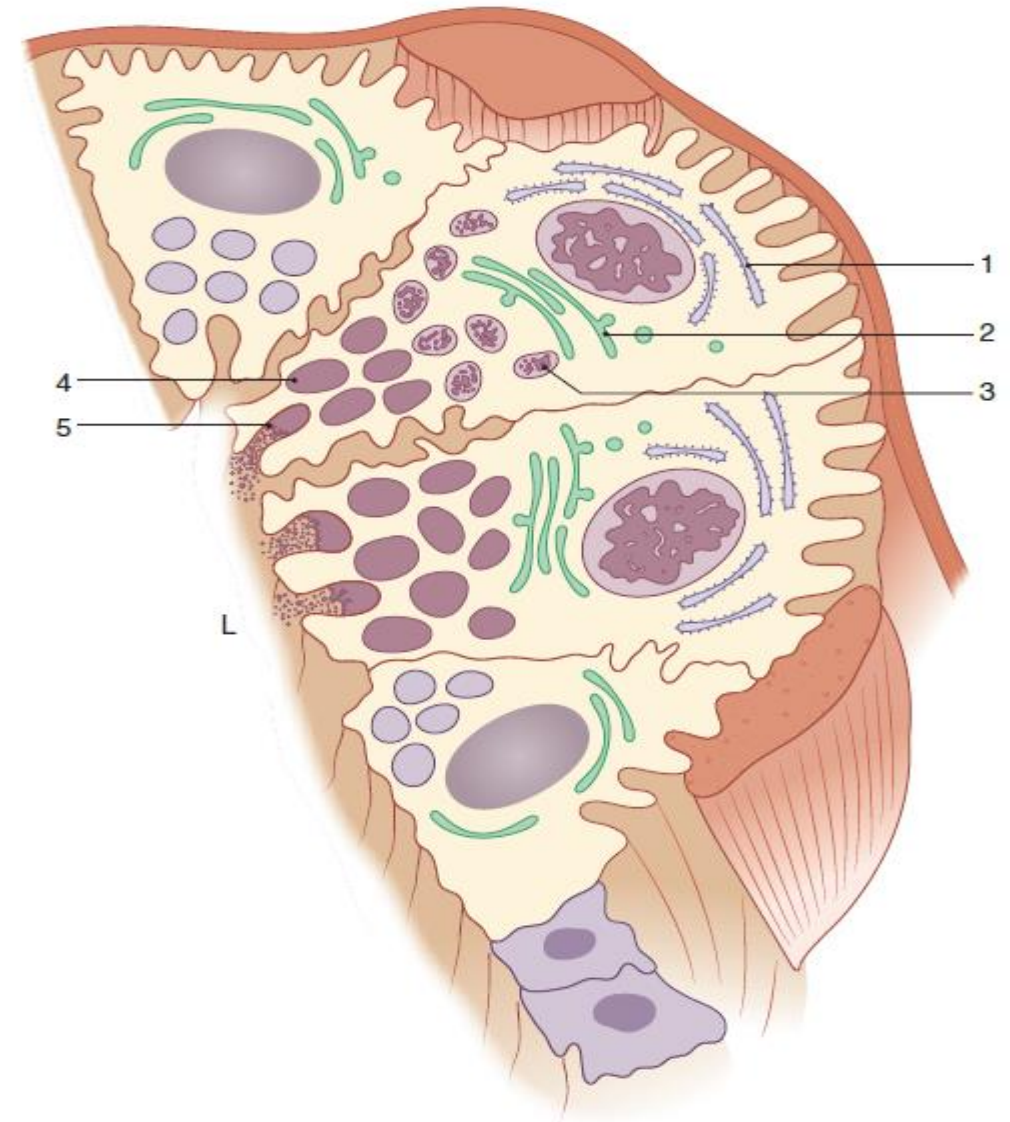


Fig. 16.3 The pathway in secretory protein synthesis, storage and stimulated exocytosis. 1 = protein synthesis in the rough endoplasmic reticulum; 2 = transport via the Golgi apparatus to the *trans*-face; 3 = initial storage within immature granules; 4 = concentration of secretory proteins in mature storage granules; 5 = exocytosis of stored proteins into the lumen of the acinus; L = lumen surface. Courtesy Prof. G B Proctor and Karger Press.

METHODS OF SALIVARY SECRETION

2. Constitutive pathway

- Proteins are secreted continuously without storage.
- Vesicles move directly from the Golgi apparatus to the plasma membrane.
- Some proteins may enter saliva, while others may pass into the blood.

Additional method: Transcytosis

- Movement of substances such as IgA across the cell from the basolateral to the apical surface.
- Helps transport immune components into saliva.

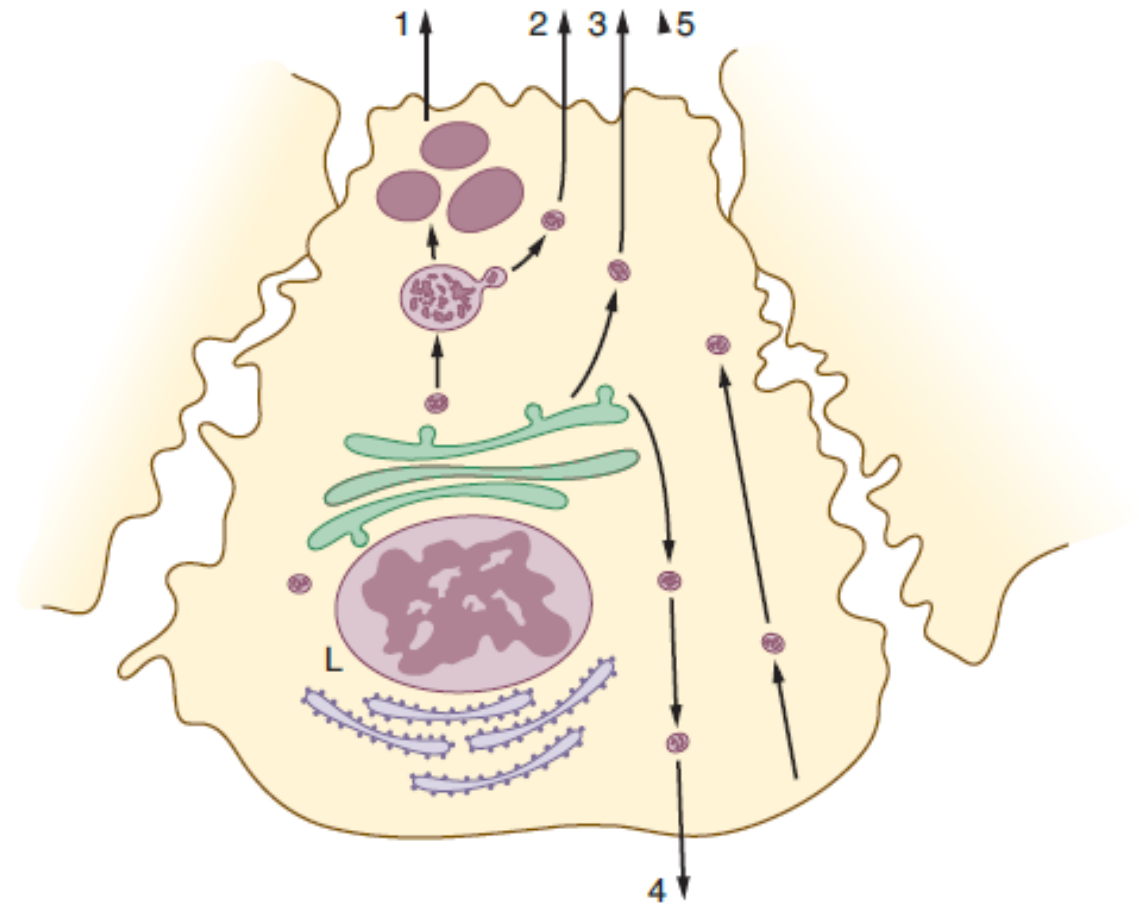
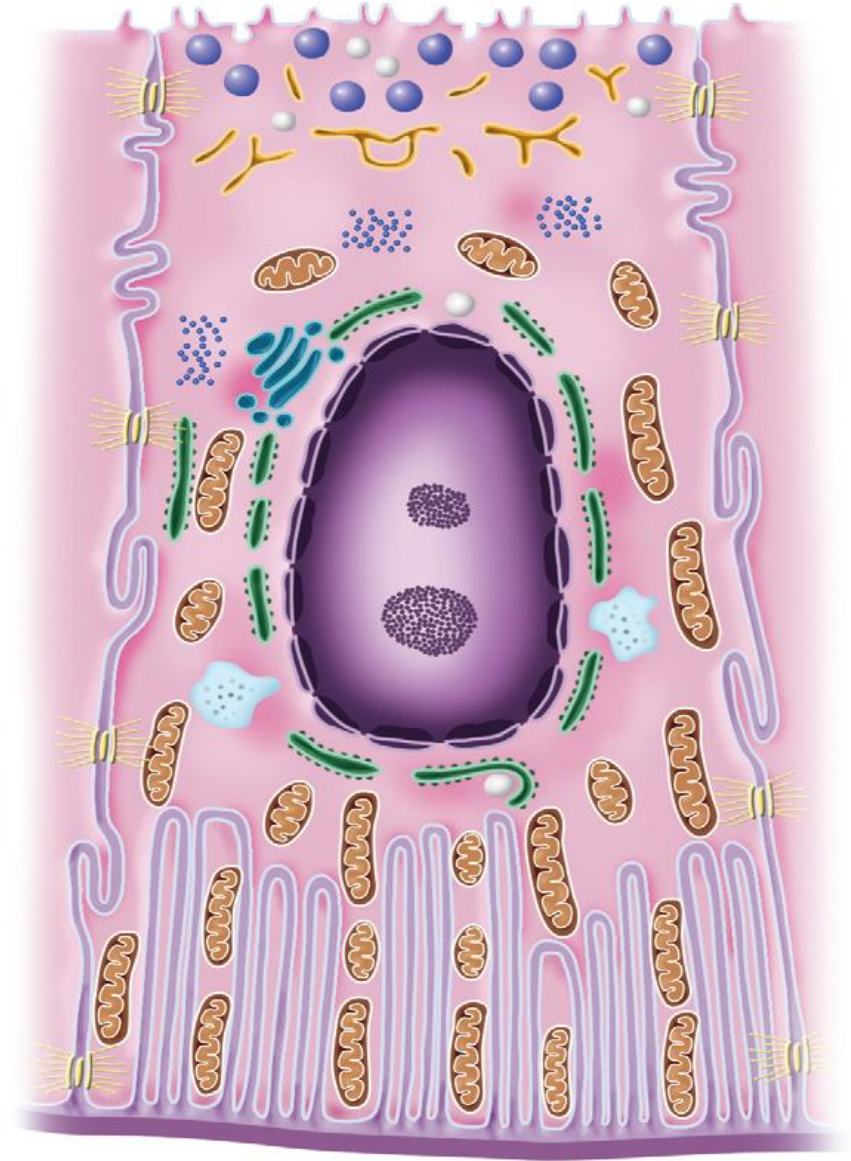


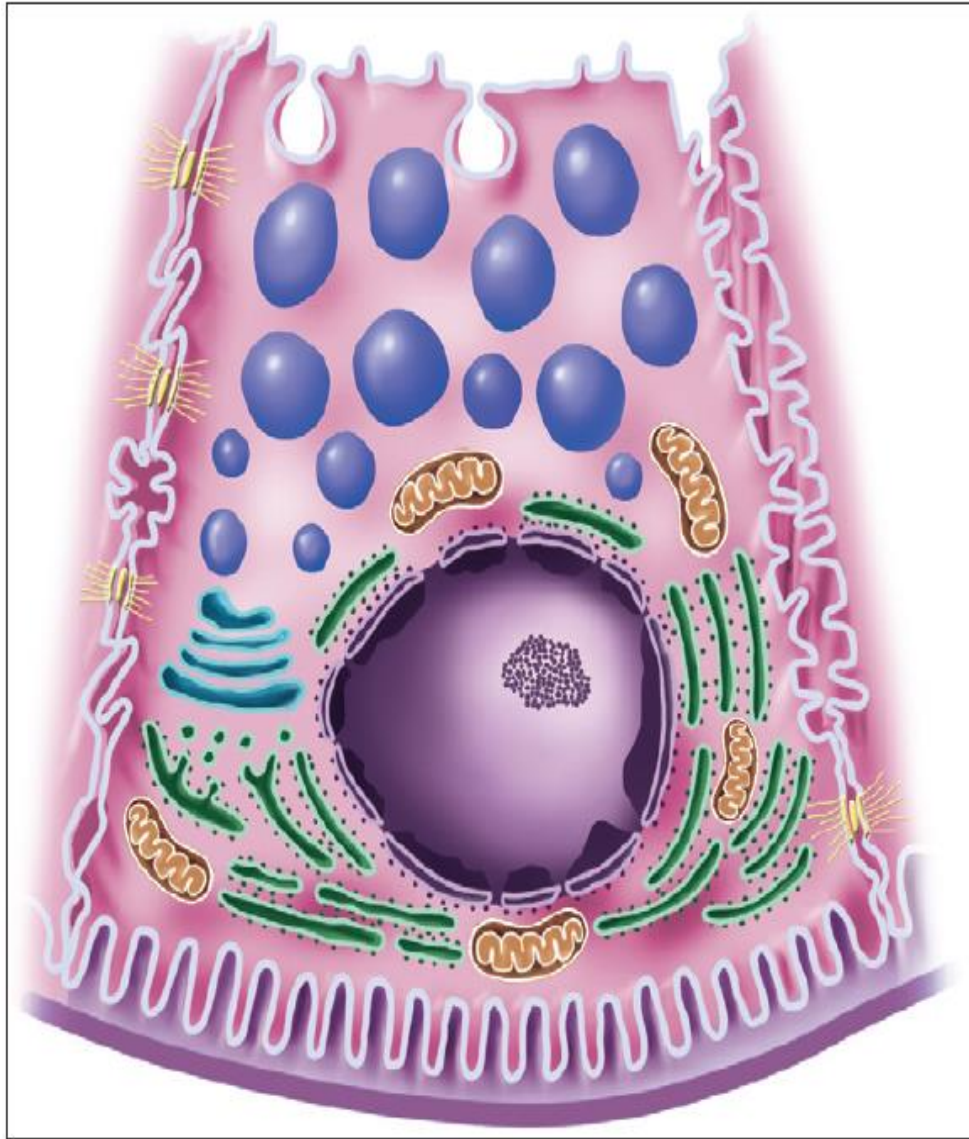
Fig. 16.4 The vesicular protein secretory pathways likely to be operating in parotid acinar cells. 1 = storage granule pathway; 2 = constitutive-like pathway; 3 = constitutive pathway to the apical membrane; 4 = constitutive pathway to the basolateral membrane; 5 = transcytosis from the basolateral to the apical membrane. L = nucleus. Courtesy Prof. G B Proctor and the Karger Press.



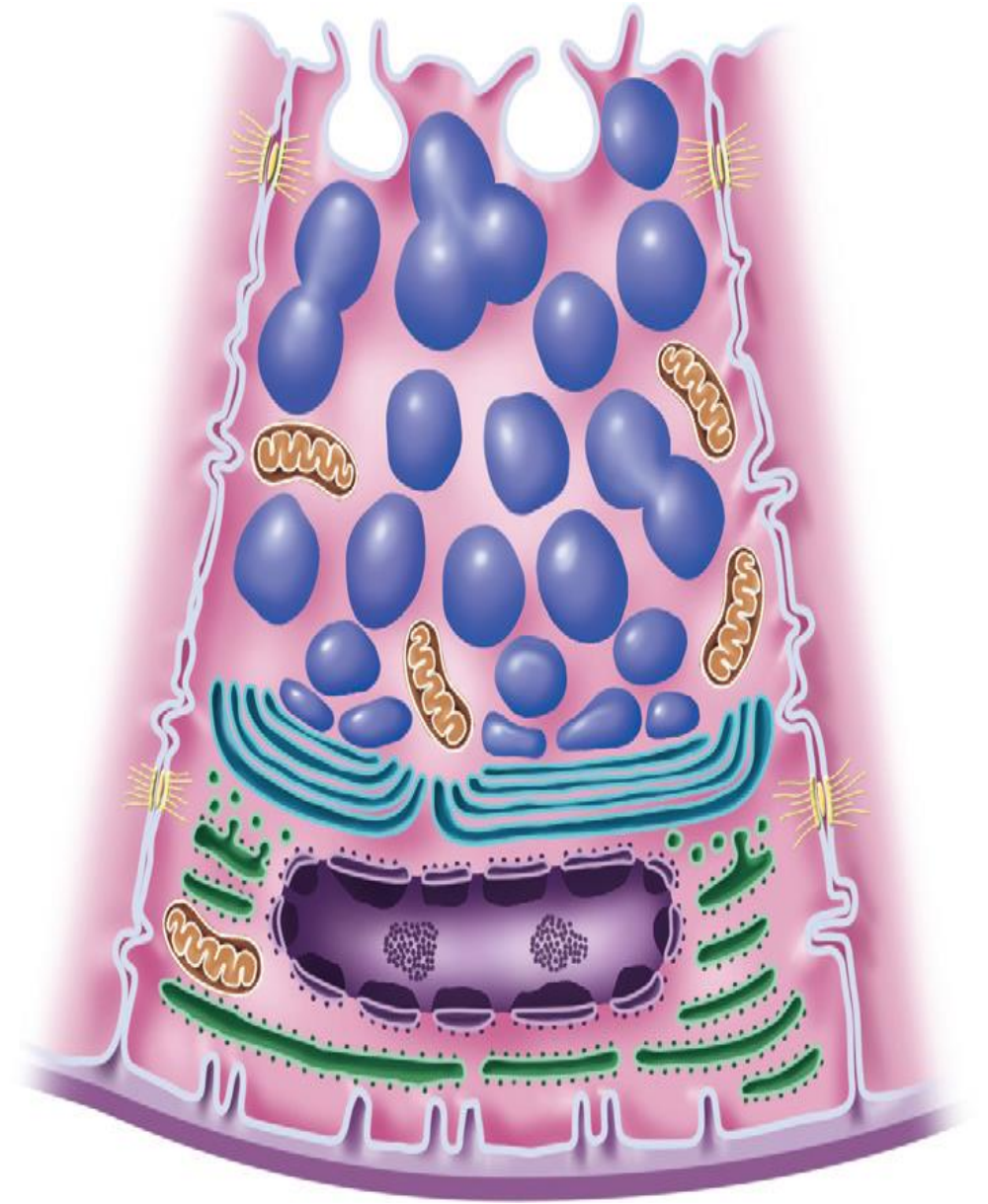
Intercalated duct cell



Striated duct cell



Serous cell



Mucous cell



Fig. 16.47 Extravasation mucocoele on the lower lip. Courtesy Dr J D Harrison.

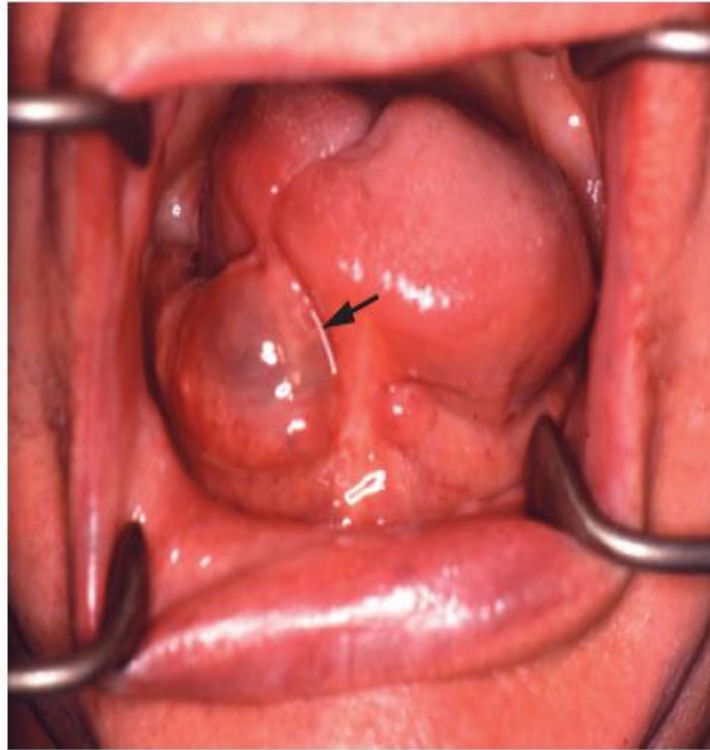


Fig. 16.48 Ranula (arrowed) in floor of mouth caused by trauma that damaged the floor of the mouth and also the tongue. Courtesy Dr J D Harrison.



Fig. 16.49 Swelling in the right neck known clinically as a plunging ranula and caused by a damaged sublingual gland from which extravasated saliva is passing through a mylohyoid hiatus. Courtesy Prof. R P Morton and the editor of the *European Archives of Oto-Rhino-Laryngology*. From Samant S, Morton RP, Ahmad Z 2011 Surgery for plunging ranula: the lesson not yet learned? *European Archives of Oto-Rhino-Laryngology* 268: 1513–1518.



Fig. 16.53 Swelling in the neck produced by enlargement of the submandibular gland as a result of chronic obstructive submandibular sialadenitis. Courtesy Dr J D Harrison.

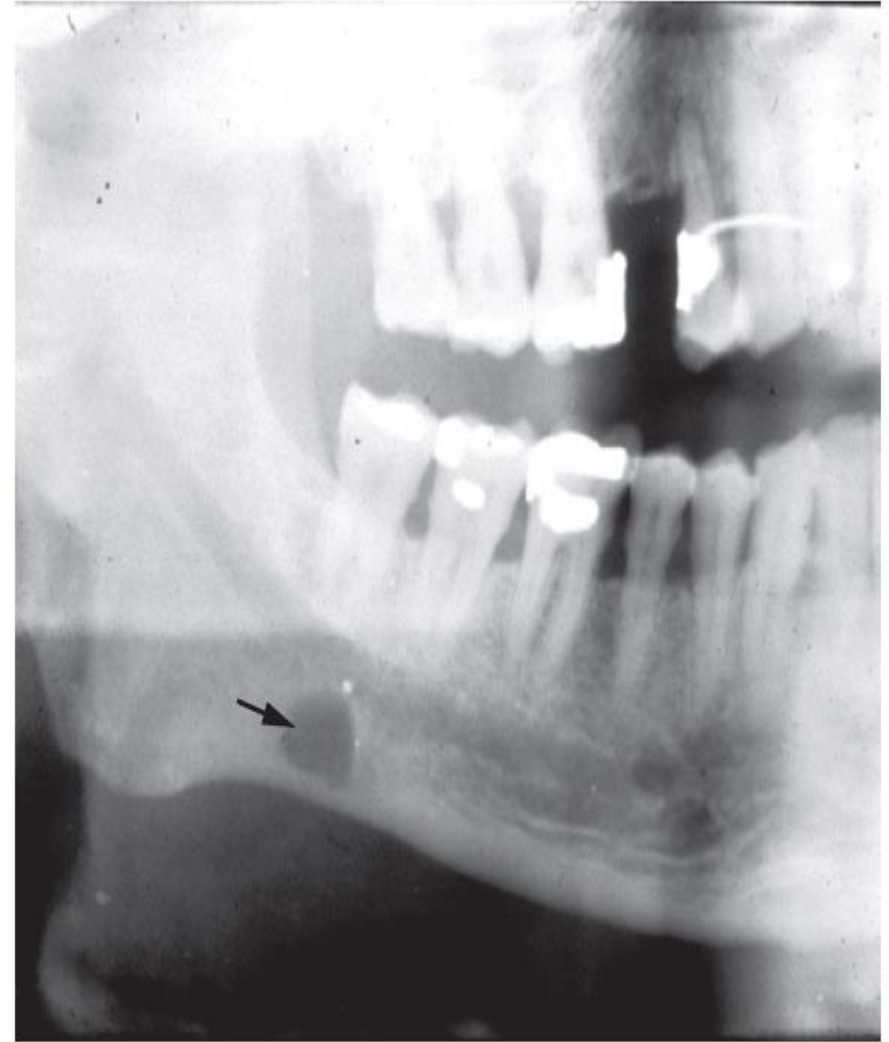


Fig. 16.45 Radiograph of a Staphne's cavity near the angle of the ramus of the mandible (arrowed). Courtesy Prof. N J D Smith.